



Cambridge International AS & A Level

 CANDIDATE
NAME


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FURTHER MATHEMATICS

9231/22

Paper 2 Further Pure Mathematics 2

May/June 2026
2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined pages at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

 This document has **16** pages.

- [illegible]



(a) Show that, for $n \geq 2$, $\pi^2 I_n = \pi(2^n + 1) - n(n-1)I_{n-2}$. [4]

This image shows a full page of primary-ruled paper. It features multiple sets of horizontal dashed lines for writing. Diagonal dotted lines are also present, creating slanted columns across the page. The background is white, and there are faint, repeating watermarks of "PapaCambridge.com" diagonally across the page.

(b) Find the exact value of I_7 . [3]

[illegible]

This image shows a full page of primary-ruled paper. It features multiple sets of horizontal dotted lines, each set separated by a larger gap. A diagonal dashed line runs from the bottom-left towards the top-right, bisecting each set of horizontal lines. The entire page is covered by a faint, repeating watermark that reads "PapaCambridge.com".


$$\frac{dy}{dx} + y \coth x = \sinh^2 x,$$

This image shows a blank sheet of white paper with horizontal ruling lines. There are three solid black lines per row, with a dashed black line centered between them, creating four equal-width bands. This pattern repeats down the entire page. Faint, light gray watermarks of the text "PapaCambridge.com" are visible diagonally across the page, repeated multiple times.

$$(x-y)^5 + \ln(xy) = 1 + \ln 2.$$

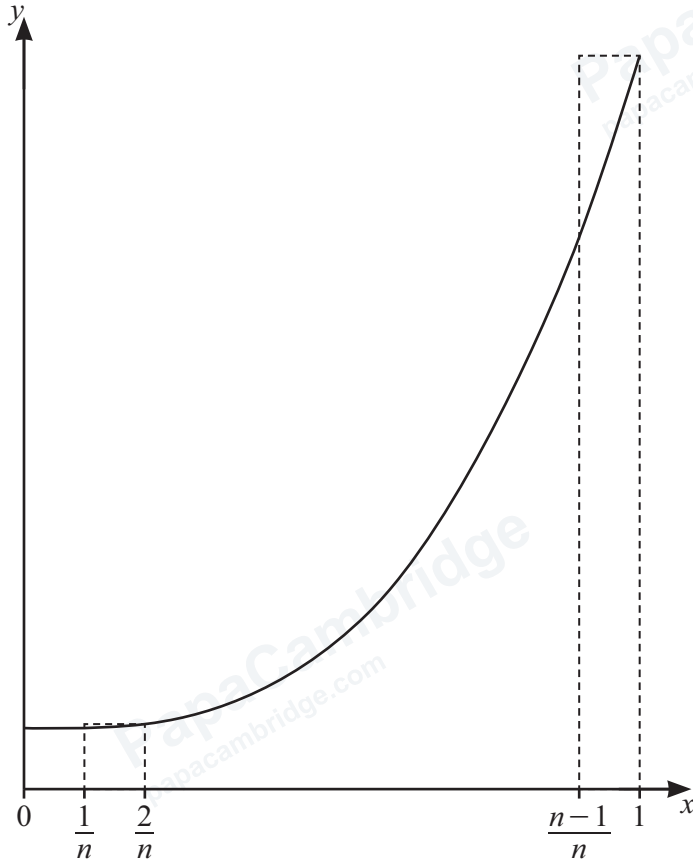
- (a) Show that, at the point $(-1, -2)$ on C , $\frac{dy}{dx} = \frac{8}{11}$. [3]

[illegible]



[5]

[illegible]



The diagram shows the curve with equation $y = \frac{11}{10}x^3 + \frac{1}{10}$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

- (a) By considering the sum of the areas of these rectangles, show that $\int_0^1 \left(\frac{11}{10}x^3 + \frac{1}{10} \right) dx < U_n$, where

$$U_n = \frac{11}{40} \left(1 + \frac{1}{n} \right)^2 + \frac{1}{10}. \quad [4]$$

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- (b) Use a similar method to find, in terms of n , a lower bound L_n for $\int_0^1 \left(\frac{11}{10}x^3 + \frac{1}{10}\right)dx$. [4]

- (c) Show that $\lim_{n \rightarrow \infty} (U_n - L_n) = 0$. [2]



$$5 \frac{d^2y}{dx^2} + 4 \frac{dy}{dx} + y = x^2 + 7x + 1,$$

given that, when $x = 0$, $y = 0$ and $\frac{dy}{dx} = 0$.

[10]



DO NOT WRITE IN THIS MARGIN

Handwriting practice area with 20 sets of dotted lines.



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$$\sum_{m=1}^{\infty} e^{-m} \cos m\theta = \frac{e \cos \theta - 1}{e^2 - 2e \cos \theta + 1}. \quad [7]$$



8 (a) The constants a and b are such that the system of equations

$$-x-2y+z=1,$$

$$ax + y - z = 2,$$

$$by + 2z = 3$$

does not have a unique solution. Find a in terms of b .

[3]

The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} -1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{pmatrix}.$$

(b) Use the characteristic equation of $\mathbf{A}$ to show that $\mathbf{A}^4 = p\mathbf{A}^2 + q\mathbf{I}$, where p and q are integers to be determined.

[4]





[7]

[illegible]

[illegible]

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